

Prescribing finite-window reaction-time modes with a static fixed-budget Doi reactivity field

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(Dated: August 14, 2026)

We ask whether a fixed amount of static spatial reactivity can prescribe reaction-time modes. For a two-particle Doi process, we prove that for every fixed finite $d \geq 2$ and $m \geq 1$ there is a nonnegative lab-frame reactivity field of fixed centre-coordinate mass $\int \kappa(c) dc = B$ for which the resulting reaction-time density has exactly m nondegenerate maxima and $m - 1$ nondegenerate minima on a declared compact positive-time window. Ordered narrow slabs along a monotone mean trajectory generate separated Gaussian exposure clocks. A global derivative-zero bound and whole-window sign certificate establish this signature uniformly over a compact family of positive slab weights; mixed-jet convergence on positive times transfers it to the killed process for sufficiently small positive B . Off-lattice simulations realize the construction at moderate noise and order-one event mass, map noise-budget mode-retention boundaries, and show predictable loss by peak merger or depletion. The result does not claim novelty for multimodal first-passage laws, topology outside the chosen window, or parameters uniform in d and m . The contribution is instead a constructive finite-window topology theorem under a static nonnegative fixed- L^1 reactivity constraint, supported by a finite-parameter realization.

I. INTRODUCTION

Reaction-time multimodality is established rather than new. Strategically arranged static traps can generate multimodal capture-time densities [1], while distinct transport channels, heterogeneous reaction control, and encounter geometries can create separated time scales [2–4]. The question here is therefore not whether several peaks can occur, but whether their complete finite-window stationary-point topology can be constructed under a specific static resource constraint.

Several adjacent inverse-design paradigms clarify that constraint. Classical inverse first-passage theory uses a time-dependent absorbing boundary to realize a prescribed survival law [5]. Soft-killing variants translate a killing profile by a time-dependent barrier and establish existence or uniqueness for prescribed lifetime distributions [6–8]. Finite-state phase-type and hyper-Erlang constructions generate flexible positive waiting-time laws by introducing hidden Markov phases [9, 10]. Stochastic gating modifies detection times through switching reaction states [11, 12], and kinetic proofreading uses sequential, energy-consuming reaction steps to impose temporal delays [13–15]. These results provide broader distributional control in different design spaces; they do not impose the static spatial constraint considered here.

Static spatial reactivity under a fixed total resource is itself also established. Heterogeneous bulk killing has been studied with fixed total reactive strength, partially reactive network elements have been rearranged at fixed total absorbance, and position-dependent Robin reactivity has been optimized under a fixed integral to maximize

steady flux [16–18]. Thus neither static killing, spatial redistribution, nor an L^1 -type resource constraint is a novelty claim here. The objective below is instead the exhaustive transient stationary-point topology of a two-particle Doi reaction-time density, together with its whole-window positive-budget transfer.

Our admissible control is a lab-frame, time-independent, nonnegative centre-coordinate field $\kappa(c)$, multiplied by a fixed encounter-contact profile and constrained by

$$\int \kappa(c) dc = B. \quad (1)$$

Every admissible allocation uses the same installed amount B . We neither translate nor activate the field in time, introduce signed reactivity, nor add an internal gating or proofreading state. Under these restrictions we ask whether a prescribed number of reaction-time modes can nevertheless be made exact on a specified time window.

a. Main result. Fix finite $d \geq 2$, finite $m \geq 1$, and a compact interval $I = [\tau, T] \subset (0, \infty)$. We construct transport and initial data, contact geometry, and m ordered normalized catalyst slabs; these data may depend on (d, m) . There is $\varepsilon_0 > 0$ such that, for every $0 < \varepsilon < \varepsilon_0$, the free-exposure clock has exactly m nondegenerate maxima and $m - 1$ nondegenerate minima on I , uniformly for all slab weights satisfying $w_j \geq w_* > 0$ and $\sum_j w_j = 1$. For each fixed such ε , there is a positive uniform topological threshold $B_{\text{top}}^{\text{unif}}(\varepsilon)$ such that the positive-budget Doi reaction-time density has the identical stationary signature whenever $0 < B < B_{\text{top}}^{\text{unif}}(\varepsilon)$. For every allocation the field remains static and satisfies Eq. (1).

The upper bound of m modes for an m -component univariate Gaussian mixture is classical [19, 20]; it is not the contribution. The proof burden created by the encounter process is to construct all $2m - 1$ stationary points with uniform margins, prove a global derivative-zero bound,

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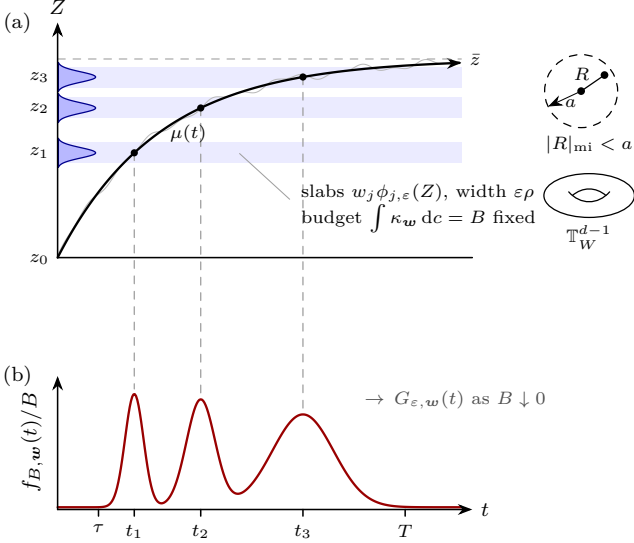


FIG. 1. Mechanism of prescribed finite modality ($m = 3$ shown). (a) The midpoint mean $\mu(t) = \bar{z} + (z_0 - \bar{z})e^{-\gamma t}$ drifts monotonically toward $\bar{z}$ (thick curve; a sample path of Z_t in gray) past three narrow normalized Gaussian catalyst slabs $w_j \phi_{j,\varepsilon}$ of width $\varepsilon\rho$ centred at the physical positions $z_j = \mu(t_j) = \text{sgn}(\mu')\ell_0 c_j$ on the Z axis, under the conserved installed budget $\int \kappa_w dc = B$ [Eq. (3)]. The deterministic relative trajectory has a strict margin inside the contact tube; the stochastic R_t may leave and re-enter it, with transverse directions on the torus $\mathbb{T}_W^{d-1}$ (right). (b) In the theorem regime $0 < B < B_{\text{top}}^{\text{unif}}(\varepsilon)$, each slab acts as a separated exposure clock: the reaction-time density develops one peak near each passage time t_j (dashed guides), and one valley between adjacent peaks, so $f_{B,w}/B$ has exactly m maxima and $m - 1$ minima on $[\tau, T]$ [Theorem 1]. (Diagram code revised with Codex assistance; scientific content author-verified.)

exclude additional zeros over the whole window after multiplication by the contact probability, and preserve the root tubes, curvature signs, endpoint slopes, and complement margin under positive-budget Doi killing. Compact-positive-time mixed-jet convergence provides the last transfer.

The theorem has a sequential quantifier order and does not supply a common useful budget, a lower event-mass bound, or parameters uniform in d and m . The off-lattice campaigns therefore serve a separate purpose: they test finite-parameter realizability, map the operational noise-budget retention boundary, and identify the predicted peak-merger and depletion failures. Their classifier is fixed in advance but operational, and no claim is made about stationary points outside I . Doi killing, reaction mass balance, and spatially heterogeneous reactivity are established ingredients [21–25].

II. CONSERVED-REACTIVITY ENCOUNTER MODEL

Let $q(x, t)$ be the unreacted configuration density of two mobile particles, with conservative generator $\mathcal{L}$. Write r for the relative coordinate and c for an encounter-centre coordinate. For a fixed contact profile $\chi_a(r)$ and normalized nonnegative centre-space profiles $\phi_j(c)$, define

$$\begin{aligned} \kappa_w(c) &= B \sum_{j=1}^J w_j \phi_j(c), & w_j &\geq 0, \\ \sum_j w_j &= 1, & \int \phi_j(c) dc &= 1, \\ K_{B,w}(x) &= \chi_a(r) \kappa_w(c). \end{aligned} \quad (2)$$

Every admissible allocation has the same installed centre-space resource,

$$\int \kappa_w(c) dc = B. \quad (3)$$

This is not an integral over the complete two-particle configuration space. The killed evolution, reaction-time density, and survival probability are

$$\begin{aligned} \partial_t q_{B,w} &= (\mathcal{L} - K_{B,w}) q_{B,w}, & q_{B,w}(0) &= q_0, \\ f_{B,w}(t) &= \langle K_{B,w}, q_{B,w}(t) \rangle, & -S'_{B,w}(t) &= f_{B,w}(t). \end{aligned} \quad (4)$$

Equation (4) is the usual mass balance for a Doi volume-reaction process, not a novelty claim.

For longitudinal catalyst slabs, translation invariance of the common transverse coordinate gives an exact quotient

$$\Omega_\infty = \mathbb{R}_z \times \mathbb{R}_{r_\parallel} \times \mathbb{T}_W^{d-1}. \quad (5)$$

In the construction below the midpoint Z_t moves monotonically in mean, the relative process remains a positive distance inside the contact region along its deterministic trajectory on the chosen window, and each normalized slab is centred at the midpoint mean at a target time. The $W^{-(d-1)}$ factor in the killing field accounts for the omitted common transverse directions and preserves Eq. (3).

III. POSITIVE-TIME TRANSFER

Write $K_{B,w} = BV_w$ and introduce interior affine control coordinates $w(\theta) = \bar{w} + P\theta$. The budget-rescaled reaction density and the free-exposure law are

$$F_B(t, \theta) = B^{-1} f_{B,w(\theta)}(t), \quad G(t, \theta) = \langle V_{w(\theta)}, e^{t\mathcal{L}} q_0 \rangle. \quad (6)$$

On either a bounded reflecting/periodic quotient or the reversible natural-decay realization of Eq. (5), assume q_0 belongs to the corresponding state space and V_w is bounded on a complex control neighbourhood of a compact interior control set. A Dyson expansion followed by

time and control Cauchy estimates gives, for every fixed time order r and finite control multi-index α ,

$$\sup_{(t,\theta) \in [\tau, T] \times \Theta} |\partial_t^r \partial_\theta^\alpha (F_B - G)| \leq C_{r,\alpha,\tau,T,\Theta} B. \quad (7)$$

The Supplemental Material accompanying this article [26] gives the explicit constant and proof. In particular, C^2 convergence on a compact positive-time window preserves disjoint stationary-point neighbourhoods, curvature signs, endpoint slopes, and a nonzero derivative margin on their complement. The estimate is local in positive time and provides neither a useful numerical value of B nor control on the scale $t = O(B^{-1})$.

A. Prescribed finite modality from separated exposure clocks

The conserved-budget support construction admits encounter processes with any prescribed *fixed finite* number of reaction-time modes. We state this existence result in an exact quotient of a two-particle Doi process; the construction depends on the prescribed dimension and mode count and does not assert mode-count switching by weight redistribution on one fixed support.

Fix finite $d \geq 2$, finite $m \geq 1$, a transverse period $W > 0$, a contact radius $0 < a < W/2$, positive scales $\ell_0, \rho > 0$, and a compact interval $I = [\tau, T] \subset (0, \infty)$. On $\mathbb{R} \times \mathbb{R} \times \mathbb{T}_W^{d-1}$, consider the independent midpoint and relative processes

$$\begin{aligned} dZ_t &= -\gamma(Z_t - \bar{z}) dt + \varepsilon \sqrt{D_0} dW_t, \\ dR_{\parallel,t} &= -\gamma R_{\parallel,t} dt + 2\varepsilon \sqrt{D_0} dW_{\parallel,t}, \\ dR_{\perp,t} &= 2\varepsilon \sqrt{D_0} dW_{\perp,t} \pmod{W}. \end{aligned} \quad (8)$$

The mutually independent initial laws satisfy

$$\begin{aligned} Z_0 &\sim N(z_0, \varepsilon^2 D_0 / (2\gamma)), \\ R_{\parallel,0} &\sim N(r_{\parallel,0}, \varepsilon^2 u_0^2), \\ R_{\perp,0} &\sim \text{WN}(r_{\perp,0}, \varepsilon^2 \Sigma_{\perp,0}), \end{aligned} \quad \begin{aligned} D_0, \gamma, u_0 &> 0, \quad z_0 \neq \bar{z}, \\ u_0^2 &< 4D_0/\gamma, \quad \Sigma_{\perp,0} \succ 0. \end{aligned} \quad (9)$$

Here WN is the wrapped Gaussian on $\mathbb{T}_W^{d-1}$, $\Sigma_{\perp,0}$ is a positive-definite $(d-1) \times (d-1)$ covariance matrix, and $|r|_{\text{mi}}$ is the Euclidean norm of the longitudinal relative coordinate together with the shortest transverse representatives modulo W .

All driving Brownian motions are also independent. Thus $\mu(t) = \bar{z} + (z_0 - \bar{z})e^{-\gamma t}$ is strictly monotone and $\text{Var } Z_t = \varepsilon^2 D_0 / (2\gamma)$. Under the minimum-image convention, assume the deterministic relative trajectory $r_*(t) = (r_{\parallel,0}e^{-\gamma t}, r_{\perp,0})$ obeys $\sup_{t \in I} |r_*(t)|_{\text{mi}} \leq a - \eta$ for some $\eta > 0$.

Choose target times

$$\begin{aligned} \tau &< t_1 < \dots < t_m < T, \quad c_j = x(t_j), \\ x(t) &= \frac{\text{sgn}(\mu')\mu(t)}{\ell_0}, \end{aligned} \quad (10)$$

so that $x'(t)$ is bounded away from zero. At the corresponding physical centres, place normalized Gaussian catalyst slabs of width $\varepsilon\rho$. Their nonnegative weights belong to a compact subset of the simplex,

$$\mathcal{W}_{w_*} = \left\{ \mathbf{w} : \sum_{j=1}^m w_j = 1, \quad w_j \geq w_* > 0 \right\}, \quad 0 < w_* \leq \frac{1}{m}, \quad (11)$$

and define

$$\begin{aligned} \phi_{j,\varepsilon}(z) &= \frac{\exp[-(z - \mu(t_j))^2 / (2\varepsilon^2 \rho^2)]}{\sqrt{2\pi} \varepsilon \rho}, \\ K_{B,\mathbf{w},\varepsilon}(Z, R) &= \mathbf{1}_{\{|R|_{\text{mi}} < a\}} \frac{B}{W^{d-1}} \sum_{j=1}^m w_j \phi_{j,\varepsilon}(Z). \end{aligned} \quad (12)$$

The slab normalization makes B the same installed centre-space budget for every allocation. Set $K_{B,\mathbf{w},\varepsilon} = BV_{\mathbf{w},\varepsilon}$, and let $T_0(t)$ and q_0 denote the unkilld semigroup and initial law. Then

$$\begin{aligned} G_{\varepsilon,\mathbf{w}}(t) &= B^{-1} \mathbb{E}_0[K_{B,\mathbf{w},\varepsilon}(Z_t, R_t)] = \langle V_{\mathbf{w},\varepsilon}, T_0(t)q_0 \rangle, \\ f_{B,\varepsilon,\mathbf{w}}(t) &= \mathbb{E}_0 \left[K_{B,\mathbf{w},\varepsilon}(Z_t, R_t) e^{-\int_0^t K_{B,\mathbf{w},\varepsilon}(Z_s, R_s) ds} \right]. \end{aligned} \quad (13)$$

The expectation $\mathbb{E}_0$ is with respect to the unkilld process started from Eq. (9). Independence gives the exact factorization

$$\begin{aligned} G_{\varepsilon,\mathbf{w}}(t) &= A_\varepsilon(t) H_{\sigma,\mathbf{w}}(x(t)), \\ A_\varepsilon(t) &= \frac{c_{d,\varepsilon}(t)}{W^{d-1} \sqrt{2\pi} \varepsilon \sqrt{D_0/(2\gamma)} + \rho^2}, \end{aligned} \quad (14)$$

where $c_{d,\varepsilon}(t) = \mathbb{P}_0(|R_t|_{\text{mi}} < a)$ and $H_{\sigma,\mathbf{w}}$ is defined below.

Theorem 1 (exact prescribed finite modality).— Fix finite $d \geq 2$, finite $m \geq 1$, the data above, and the compact window I . There is $\varepsilon_0 > 0$ such that, for every $0 < \varepsilon < \varepsilon_0$ and every $\mathbf{w} \in \mathcal{W}_{w_*}$, the unit-budget free-exposure clock has exactly m nondegenerate maxima and $m-1$ nondegenerate minima on I , ordered alternately, and no stationary endpoint. For each fixed such ε and allocation set $\mathcal{A} \subseteq \mathcal{W}_{w_*}$, define the connected topology-retention budget

$$\begin{aligned} B_{\text{top}}(\varepsilon; \mathcal{A}) &:= \sup \{ b > 0 : \\ &\quad f_{\beta,\varepsilon,\mathbf{w}}/\beta \text{ has the stated} \\ &\quad \text{complete stationary signature on } I \\ &\quad \text{for every } \mathbf{w} \in \mathcal{A} \text{ and every } 0 < \beta < b \}. \end{aligned} \quad (15)$$

Then

$$B_{\text{top}}^{\text{unif}}(\varepsilon) := B_{\text{top}}(\varepsilon; \mathcal{W}_{w_*}) > 0. \quad (16)$$

Consequently the budget-rescaled Doi reaction-time density has the stated signature for every $0 < B < B_{\text{top}}^{\text{unif}}(\varepsilon)$ and every admissible allocation. Multiplication by B leaves the stationary points unchanged, so the conclusion also holds for the physical density $f_{B,\varepsilon,w}$.

The mechanism can be seen without invoking a normal form. Gaussian convolution reduces the longitudinal part of the unit-budget exposure to

$$H_{\sigma,w}(x) = \sum_{j=1}^m w_j \exp\left[-\frac{(x - c_j)^2}{2\sigma^2}\right], \quad (17)$$

$$\sigma = \frac{\varepsilon}{\ell_0} \sqrt{\frac{D_0}{2\gamma} + \rho^2}.$$

Writing $q_j(x) = w_j \exp[-(x - c_j)^2/(2\sigma^2)]$, $\pi_j(x) = q_j(x)/H_{\sigma,w}(x)$, and $L = \partial_x \log H$, let $\bar{c}(x)$ and $\text{Var}_{\pi(x)}(c)$ denote the mean and variance of the centres under the finite probability vector $\pi(x)$. Then

$$L(x) = \frac{\bar{c}(x) - x}{\sigma^2}, \quad L'(x) = \frac{\text{Var}_{\pi(x)}(c)}{\sigma^4} - \frac{1}{\sigma^2}. \quad (18)$$

The univariate upper bound of m modes for an m -component Gaussian mixture is classical [19, 20]. Here a direct affine-exponential proof is retained because the Doi transfer needs all critical zeros counted with multiplicity together with uniform root-tube, complement, curvature, and endpoint margins.

For the adjacent pair $c_j < c_{j+1}$, the weighted crossover is

$$s_j = \frac{c_j + c_{j+1}}{2} + \frac{\sigma^2}{c_{j+1} - c_j} \log \frac{w_j}{w_{j+1}}. \quad (19)$$

The adjacent ratio obeys $q_{j+1}/q_j = 1/9$ at $s_j - \sigma^2 \log(9)/(c_{j+1} - c_j)$ and $q_{j+1}/q_j = 9$ at the corresponding plus edge. Thus both adjacent components remain uniformly present throughout the crossover, while all nonadjacent components are exponentially small. Equation (18) then yields one simple minimum at

$$r_j = s_j + O(\sigma^4), \quad L'(r_j) = \frac{(c_{j+1} - c_j)^2}{4\sigma^4} + O(\sigma^{-2}). \quad (20)$$

Each centre similarly contributes one simple maximum at $p_j = c_j + O(\exp[-q/\sigma^2])$, with $L'(p_j) = -\sigma^{-2} + o(\sigma^{-2})$. These constructions give $2m - 1$ simple stationary points. They are exhaustive: after multiplication by a positive Gaussian factor, H' is an exponential polynomial

$$\sum_{j=1}^m (a_j + b_j x) e^{\lambda_j x}, \quad \lambda_1 < \dots < \lambda_m, \quad (21)$$

which has at most $2m - 1$ real zeros counted with multiplicity. The extreme affine-exponential terms dominate in the two tails, so all zeros lie in a compact interval and real analyticity makes their number finite. A generalized Rolle induction then proves the bound by differentiating twice to remove the lowest-exponent affine term.

The remaining relative-coordinate contact probability is a positive slow factor. Uniform bounds on its first two logarithmic time derivatives, peak boxes of width $O(\sigma^2)$, valley boxes of width $O(\sigma^4)$, and a posterior-sector sign certificate exclude stationary points on the full box complement. Consequently the slow factor moves maxima by $O(\sigma^2)$ and minima by $O(\sigma^4)$ without changing their number or type. Finally, on every fixed positive ε , compact-positive-time mixed-jet convergence

$$\sup_{w \in \mathcal{W}_{w_*}} \left\| \frac{f_{B,\varepsilon,w}}{B} - G_{\varepsilon,w} \right\|_{C^2(I)} \rightarrow 0 \quad (B \downarrow 0) \quad (22)$$

preserves the disjoint root tubes, signed curvature margins, complement margin, and endpoint slopes.

The quantifier order is essential: the geometry and admissible scales may depend on fixed (d, m) ; one first fixes a sufficiently small $\varepsilon > 0$ and only then chooses $B < B_{\text{top}}^{\text{unif}}(\varepsilon)$. The theorem does not provide a useful lower bound for that threshold, a uniform-in- d or uniform-in- m construction, an event-mass floor, or a stationary-point count outside I ; it also uses a whole-window contact-interior condition whose small-noise limit is asymptotically saturated contact. Nontrivial contact, observability at a common positive budget, and finite-parameter robustness remain separate numerical checks.

Three thresholds are kept distinct. The theorem concerns the maximal uniform topology-preservation threshold $B_{\text{top}}^{\text{unif}}(\varepsilon)$. For one fixed allocation, the Supplemental Material [26] derives a closed-form sufficient value $B_{\text{cert}}(\varepsilon; w, \{U_i\})$, from a flat Dyson-chain estimate, such that every $0 < B < B_{\text{cert}}$ preserves the free-clock signature. Only this sufficient value, along the fixed anchor family used in its evaluation, has the established asymptotic form

$$\log B_{\text{cert}} = -\frac{c_B}{\varepsilon^2} + O\left(\log \frac{1}{\varepsilon}\right), \quad c_B > 0. \quad (23)$$

Here $\{U_i\}$ is the selected system of disjoint root tubes; this dependence is suppressed below. The construction proves only $B_{\text{cert}} \leq B_{\text{top}}(\varepsilon; \{w\})$, not a bound for the threshold uniform over the full allocation family. High-precision, non-interval evaluations using the same relative-coordinate initial law as the simulations give nominal values $\hat{B}_{\text{cert}} \simeq 8.9 \times 10^{-15}$ for the $m = 2$, $\varepsilon = 0.1$ anchor and 2.6×10^{-7} for the corresponding $m = 3$ geometry. By contrast, B_{op} below is a finite-sample crossing of a stated classifier: $B_{\text{op}} \simeq 7.6$ and 3.6 , respectively. It is not identified with either B_{top} or B_{cert} . The large gap is dominated by the whole-window left-endpoint derivative margin, while the weighted-norm obstruction of the direct L^2 route is absent from the flat-pairing formula.

IV. NUMERICAL REALIZATION

The theorem is an existence statement with a deliberately sequential quantifier order, and by itself does not identify a practically resolved parameter set. This section reports an off-lattice time-discrete simulation of the same continuum model at moderate noise and order-one budgets, designed to test the theorem’s falsifiable finite-parameter implications, including its predicted failure directions. Table I summarizes the construction as the checkable procedure that the campaigns below execute.

A. Method and sanity checks

No spatial lattice is introduced. The midpoint, longitudinal, and transverse coordinates of Eq. (8) are integrated by Euler–Maruyama steps of size $\Delta t = 10^{-3}$, and Doi killing is realized by per-step thinning with probability $1 - \exp(-K_{B,w,\varepsilon} \Delta t)$ evaluated at the current position. All runs use deterministically seeded counter-based random streams, one substream per walker chunk. Two automated checks accompany every configuration: the per-step kill probability must remain in the unit interval (largest value 5.6×10^{-3} across all production configurations), and event counts must balance mass exactly (kills plus survivors equal walkers). Separately, a campaign-level zero-budget twin run accumulates about 10^7 contact steps and produces no kills. A Δt -halving twin of the $(m, \varepsilon, B) = (2, 0.1, 1)$ configuration reproduces both peak times at bin resolution and both peak prominences to within 0.3%, and the $d = 3$ run passes a per-bin Δt -halving comparison with maximum $|z| = 1.67$. Modality is decided by a fixed operational classifier: reaction times are histogrammed with bin width 0.02 and smoothed with a Gaussian kernel of bandwidth 0.04, and a local maximum counts as a mode only if its topographic prominence (peak minus contour base) exceeds both five covariance-aware smoothed-Poisson standard deviations and 5% of the maximum smoothed height on the window $I = [0.5, 3.5]$. Peak and contour base are linear statistics of the raw bin counts, so the prominence variance is $\sum_j (A_{p,j} - A_{b,j})^2 C_j / (N\delta)^2$, with A the kernel smoothing weights, C_j the raw counts, N the walker count, and δ the bin width; a peak-only Poisson sigma, which omits the base variance and the peak–base covariance, would only overstate the significance ratio of counted maxima (by factors up to 1.55 across the retained campaign) and is not used. Post-hoc sensitivity analysis over bandwidths 0.03–0.05 and relative-prominence floors 3%–7% leaves the principal anchors unchanged but changes the verdict in four of the 96 phase-diagram cells adjacent to the reported boundary. Complete method details, configuration tables, retained histogram counts, and additional figures are given in the Supplemental Material [26].

a. Computational-tool disclosure. OpenAI Codex Desktop 0.147.0-alpha.6.5 (GPT-5.6-sol, August 2026), with earlier GPT-5-family Codex sessions (July–August

TABLE I. Finite-window design recipe implied by Theorem 1. Each step cites the statement that licenses it; the recipe prescribes modality only inside the declared window I .

Given finite $d \geq 2$ and $m \geq 1$, a window $I = [\tau, T] \subset (0, \infty)$, and target times $\tau < t_1 < \dots < t_m < T$:
(1) Transport. Choose $(\gamma, z_0, \bar{z})$ with $z_0 \neq \bar{z}$ so that the midpoint mean $\mu(t) = \bar{z} + (z_0 - \bar{z})e^{-\gamma t}$ traverses the design range monotonically with $x'(t)$ bounded away from zero on I [Eq. (10)]. On a compact window $x' > 0$ holds automatically; the quantitative point is to keep $T \lesssim$ a few γ^{-1} so $x'(t_m)$ is not exponentially small (late peaks widen as $\sigma/x'(t_j)$).
(2) Support. Place the slab centres at $c_j = x(t_j)$, i.e. physical centres $\mu(t_j)$ [Eq. (10)].
(3) Width. Choose ε (and ρ) so that σ of Eq. (17) obeys $\min_j (c_{j+1} - c_j) \geq C\sigma$ with $C \approx 2.3$. At this separation the crossover analysis [Eqs. (19)–(20)] yields one valley per adjacent gap with nonadjacent components exponentially small; strongly unbalanced weights shift the crossover by $\sigma^2 \log(w_j/w_{j+1})/(c_{j+1} - c_j)$ and eat into the margin. Theorem 1 guarantees only that some $\varepsilon < \varepsilon_0$ works; $C \approx 2.3$ is a practical proxy, slightly above the equal-weight two-Gaussian bimodality threshold 2σ .
(4) Verify the free clock. Evaluate $G_{\varepsilon,w}(t) = A_\varepsilon(t)H_{\sigma,w}(x(t))$ [Eqs. (14)–(17)] on I —a cheap one-dimensional computation; the contact factor $c_{d,\varepsilon}(t)$ follows from Gaussian and wrapped-Gaussian marginals—and confirm exactly m nondegenerate maxima and $m - 1$ nondegenerate minima, with clear root separation, curvature signs, and endpoint slopes.
(5) Budget. Theorem 1 guarantees the complete signature for $B < B_{\text{top}}(\varepsilon; \mathcal{W}_{w_*})$, where B_{top} is the connected topology-retention threshold uniform over the declared allocation family; equivalently, $f_{B,\varepsilon,w}/B$ inherits the complete stationary signature of $G_{\varepsilon,w}$ [Theorem 1; Eqs. (7), (22)]. The theorem asserts that this threshold is positive. For one fixed allocation, the Supplemental Material gives a sufficient expression $B_{\text{cert}}(\varepsilon; \mathbf{w}, \{U_i\}) \leq B_{\text{top}}(\varepsilon; \{\mathbf{w}\})$; its displayed decimal evaluations are high-precision floating-point values, not interval certificates. At practical scales, use the operational classifier crossing $B_{\text{op}}(\varepsilon; m, \mathbf{w}, \mathcal{P})$ only as a separate numerical gate and re-verify f directly at the chosen B (Sec. IV H).
<i>Caveat (quantifier order).</i> The choices are sequential: the geometry and admissible scales may depend on (d, m) ; one first fixes $\varepsilon < \varepsilon_0$ and only then $B < B_{\text{top}}(\varepsilon; \mathcal{W}_{w_*})$. Nothing is claimed about modality outside I , about an event-mass floor, or about uniformity of ε_0 or B_{top} in d or m (Secs. IV H and V).

2026), assisted with software and plotting-code drafting, workflow orchestration, analytical derivation and proof checks, adversarial tests, and reproducibility checks. The author set the questions, model, protocols, acceptance criteria, and claims; verified retained outputs using deterministic seeds, machine-readable histogram and classifier records, targeted reruns, and a checksummed archive; and

remains responsible for their interpretation and accuracy.

B. Production grid at $m = 2$ and $m = 3$

Eighteen production configurations with 5×10^6 walkers each realize the slab design of Theorem 1 at $d = 2$. For $m = 2$, all twelve configurations across $\varepsilon \in \{0.1, 0.2\}$, $B \in \{0.5, 1, 2\}$, and weights $(\frac{1}{2}, \frac{1}{2})$ and $(0.7, 0.3)$ are classified as bimodal under the fixed rule, with the weakest counted peak still 72 Poisson standard deviations prominent. Event mass is order one throughout: between 0.29 and 0.92 of all walkers react inside the window. The early peak sits within 0.01–0.03 of its target time in every configuration; the late peak is pulled earlier by progressive depletion as the budget grows, by 0.05–0.11 at $B = 0.5$, 0.09–0.19 at $B = 1$, and 0.15–0.29 at $B = 2$.

For $m = 3$ at $\varepsilon = 0.1$, $B = 1$, the fixed classifier identifies three significant maxima for both the uniform and the $(0.5, 0.3, 0.2)$ allocations, near the targets $(0.8, 1.6, 2.8)$, with window event mass 0.66–0.74. At $\varepsilon = 0.2$, $B = 1$, both allocations lose the third mode — consistent with the heuristic design margin becoming marginal: the minimum adjacent centre spacing equals 2.30σ at $\varepsilon = 0.2$, at the $C \approx 2.3$ spacing boundary of Table I, and the exact free clock’s third-peak prominence there is only 5.1% of the window maximum, marginally above the classifier’s 5% floor. Reducing the budget to $B = 0.25$ at uniform weights partially restores the third maximum, which is visible at $t = 2.49$ but uncertified: its prominence is 4.8σ , below the five-sigma gate, and 2.7% of the maximum height, below the 5% floor, so the strict classifier records two modes; consistently, the threshold study below certifies no third-mode crossing anywhere in the scanned budget range for this column.

C. Phase diagram and mode-retention boundary

A 96-cell campaign at equal slab weights scans $\varepsilon \in \{0.05, 0.075, 0.1, 0.125, 0.15, 0.175, 0.2, 0.25\}$ against six budgets $B \in \{0.125, 0.25, 0.5, 1, 2, 4\}$ for both designs under the stated operational rule (Fig. 2). The $m = 2$ design is classified as bimodal in all 48 cells. The $m = 3$ design retains all three classifier-significant modes at every budget for $\varepsilon = 0.05$; loses only the largest budget $B = 4$ for $\varepsilon = 0.075$ –0.125; loses $B \geq 2$ at $\varepsilon = 0.15$ and $B \geq 1$ at $\varepsilon = 0.175$; and fails at every budget for $\varepsilon = 0.2$ (the classifier-floor column) and $\varepsilon = 0.25$, where the semi-analytic free clock itself retains only two window maxima. The observed staircase is monotone in both variables, and the B -independent loss of the third free-clock maximum is bracketed to $\varepsilon \in (0.2, 0.25]$, overlaid on the figure.

D. Operational budget thresholds

For each ε , an equal-weight bisection chain measures the protocol-defined crossing $B_{\text{op}}(\varepsilon; m, \mathbf{w}_{\text{eq}}, \mathcal{P})$ at which the last mode changes its classifier verdict (Fig. 3). The two designs behave qualitatively differently. For $m = 2$ the measured thresholds are $B_{\text{op}} = 6.99$ at $\varepsilon = 0.05$ and $B_{\text{op}} = 7.62$ at $\varepsilon = 0.1$, and the threshold exceeds the scan limit 8 at $\varepsilon = 0.15$ and 0.2: a mildly increasing trend. For $m = 3$ the operational crossings fall steeply, 4.15, 3.57, and 1.55 at $\varepsilon = 0.05, 0.1$, and 0.15; at $\varepsilon = 0.2$ no probe in the scanned budget range $[0.125, 8]$ certifies a third mode, so no operational crossing exists there at 10^6 -walker resolution.

One remark is required at $(m, \varepsilon) = (3, 0.2)$. There the exact free clock’s third-peak prominence is 5.1% of the window maximum — marginally above the classifier’s 5% floor even in the zero-budget limit — so this column probes the classifier floor itself. Under the covariance-aware statistic the two independent streams agree: the phase-diagram grid records two classified modes at every budget in this column, and every threshold probe fails the five-sigma rule (the closest candidates, at $B \leq 0.28$, reach only 4.2 – 4.4σ). A peak-only sigma convention, which omits the base variance and the peak–base covariance, would instead pass three probes in this column at a nominal 5.8 – 6.1σ and would manufacture both an apparent crossing near $B = 0.28$ and an unexplained disagreement between the two streams; the covariance-aware statistic removes both. The remaining seven chains are unaffected: every bracket-defining probe verdict is unchanged under the corrected statistic. Their numerical crossings remain classifier-dependent. Across the nine post-hoc classifier settings, bracketed midpoint spans include 6.17 – 7.83 for $(m, \varepsilon) = (2, 0.1)$ and 3.08 – 3.63 for $(m, \varepsilon) = (3, 0.1)$, with some settings censored by the stored probe ranges; complete spans and censoring counts are reported in the Supplemental Material.

E. Robustness and failure modes

Beyond the theorem’s hypotheses, a jitter campaign perturbs the slab centres by independent Gaussian displacements of scale ηs (with s the minimum adjacent centre spacing) and the weights by a Dirichlet($50 \bar{\mathbf{w}}$) draw, with 50 replicas of 2×10^5 walkers per point at the $\varepsilon = 0.1$, $B = 1$ anchors. The target classifier mode count survives in at least 96% of replicas up to $\eta = 0.15$ for both designs, decaying to 46% ($m = 2$) and 40% ($m = 3$) at $\eta = 0.4$; all 50 weight-jitter replicas retain the target classifier mode count for both designs. The observed pattern is predominantly consistent with geometric merger: replica-level classifier verdicts agree with the semi-analytic verdict of each replica’s own perturbed geometry in 86–100% of cases, and the mismatches preclude a purely geometric attribution. Classified failures are associated mainly with mergers of adjacent peaks when jittered spacing collapses,

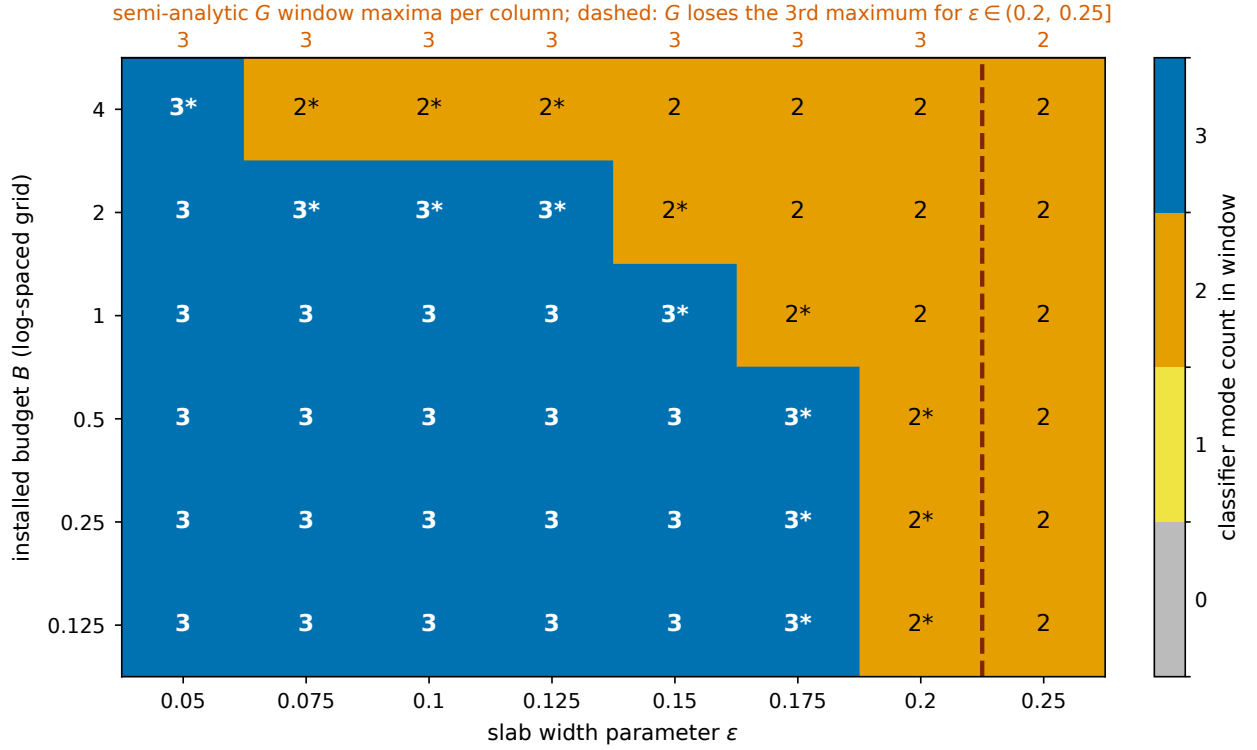


FIG. 2. Operational modality phase diagram for the $m = 3$ design: equal slab weights and classifier mode count over $\varepsilon \in \{0.05, \dots, 0.25\}$ and $B \in \{0.125, \dots, 4\}$, 10^6 walkers per cell ($*$ = refined at 3×10^6). The retained-mode boundary is a monotone staircase in (ε, B) ; the dashed line marks the B -independent semi-analytic boundary $\varepsilon \in (0.2, 0.25]$ beyond which the free exposure clock G itself loses its third window maximum. The $\varepsilon = 0.2$ column sits at the classifier floor discussed in Sec. IV D. The companion $m = 2$ diagram (Supplemental Material) is classified as bimodal under the fixed rule in all 48 cells. A classifier-sensitivity sweep moves four $m = 3$ cells adjacent to the displayed boundary; the anchor $(\varepsilon, B) = (0.1, 1)$ is stable over the tested range. (Plotting code revised with Codex assistance; numerical content author-verified.)

rather than with extra peaks.

F. Higher mode count and dimension

Two further campaigns probe the prescribed- m and prescribed- d directions at fixed design effort. For a five-slab design on a stretched geometry, the fixed classifier identifies five significant maxima near the five target times (to within 0.05), with prominences $215\text{--}454\sigma$ and window event mass 0.53 per walker (Fig. 4). A $d = 3$ spot check — two transverse torus directions, killing prefactor B/W^2 , all other parameters unchanged from the $(m, \varepsilon, B) = (2, 0.1, 1)$ anchor — is classified with two significant maxima at $t = 0.99$ and 2.37 , with window event mass 0.60 per walker (Supplemental Material). This check is consistent with the quotient construction in the first higher dimension; it is not a dimension-uniform numerical result.

G. Consistent dimensionalization of the numerical anchor

Comparison of the generator and stochastic differential equations gives the physical single-particle diffusivity $D = \varepsilon^2 D_0$, not D_0 . With length and time units $1 \mu\text{m}$ and 1s , the $\varepsilon = 0.1$, $D_0 = 1$ anchor therefore corresponds to $D = 0.01 \mu\text{m}^2\text{s}^{-1}$. Retaining the simulated $W = 1$ gives a $1 \mu\text{m}$ transverse period and, for the equal-weight $m = 3$, $B = 1$ case, an on-contact single-slab peak rate of 1.33s^{-1} . This is a dimensionalization of the simulated point, not evidence that $B = 1$ lies below B_{top} ; the full mapping is given in the Supplemental Material.

H. What the numerics establish, and what they do not

The campaigns give finite-parameter evidence that the designed signatures can be observed at moderate noise, order-one budgets, and order-one event mass. They also locate failures in the anticipated directions: depletion at large budget, clock merger as noise or centre jitter grows, and loss of a free-clock maximum when its separation

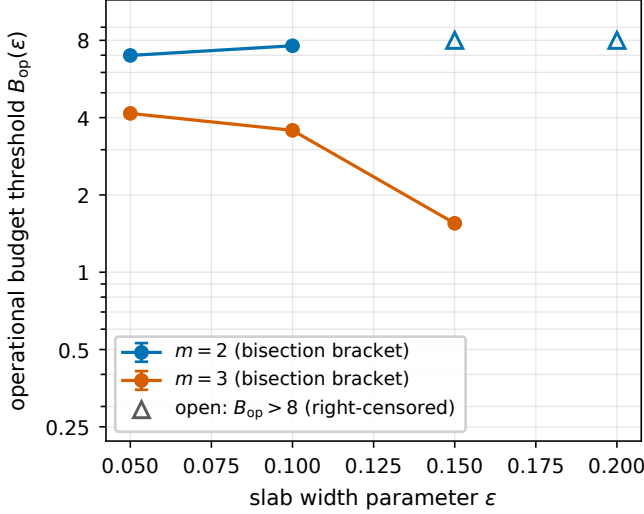


FIG. 3. Empirically bisected operational budget crossing $B_{\text{op}}(\epsilon; m, w_{\text{eq}}, \mathcal{P})$ for equal slab weights and the fixed protocol $\mathcal{P}$. The crossing is determined by the stated last-mode pass rule and reported with geometric bisection brackets at 10^6 walkers per probe. For $m = 2$ the threshold is mildly increasing in ϵ and right-censored at the scan limit $B = 8$ beyond $\epsilon = 0.1$; for $m = 3$ it falls steeply as the design margin shrinks, and at $\epsilon = 0.2$ no scanned budget certifies the third mode, so no crossing is plotted there (see text). (Plotting code revised with Codex assistance; numerical content author-verified.)

margin closes. These are classifier outcomes, not exact root counts for the time-discretized process.

No spatial lattice is introduced, but the simulations retain both time-step and sampling error. The Δt -halving comparisons are consistency checks, not a convergence theorem. Three independent seeds agree on all 18 verdicts across six targeted configurations, and the $m = 3$ and $m = 5$ interior anchors retain their mode counts when Δt is halved. However, one independent $\Delta t/2$ run of the boundary-adjacent $(m, \epsilon, B) = (3, 0.175, 0.5)$ cell changes from three to two classified modes because the late candidate peak falls below both classifier thresholds. Since the step size also changes the random-number stream, this comparison does not isolate time-step from sampling error; the displayed phase-grid verdict is resolution/stream-sensitive. Nothing numerical is uniform in (d, m) . The sufficient-bound formula B_{cert} and the operational crossing B_{op} differ by roughly fifteen decades at the matched $m = 2$ anchor and seven for $m = 3$; the displayed B_{cert} values are reported as $\hat{B}_{\text{cert}}$: high-precision evaluations without interval enclosures. Finally, the principal anchors are stable over the tested classifier ranges, whereas four of the 96 phase-diagram cells change verdict under the sensitivity sweep and the $(m, \epsilon) = (3, 0.2)$ column lies at the relative-prominence floor.

V. DISCUSSION

The theorem occupies a narrow gap between spatial first-passage theory and inverse timing design. Relative to strategically arranged absorbing traps, it provides an arbitrary prescribed fixed finite mode count, an exhaustive maxima-minima count on a declared window, uniformity over a compact family of positive allocations, and transfer to a two-particle positive-budget Doi law under a conserved centre-coordinate resource. Relative to inverse first-passage constructions, it uses a static lab-frame field but prescribes only finite-window topology rather than an arbitrary lifetime distribution. Relative to phase-type, gated, and proofreading mechanisms, it introduces no hidden reaction phase or time-dependent switch. The novelty is this conjunction of constraints, not multimodality by itself.

Mathematically, stationary catalyst locations become separated exposure clocks. The affine-exponential zero bound makes the Gaussian-clock count exhaustive; root-tube and posterior-sector estimates retain a derivative margin on the entire complement; the contact factor perturbs rather than creates the clock signature; and compact-positive-time C^2 convergence transfers all maxima, minima, and curvature signs to the killed law. Uniformity over $w_j \geq w_*$ is important: the conclusion is not tied to one finely tuned allocation, although changing weights on the same support is not claimed to switch among arbitrary mode counts.

The simulations provide finite-parameter evidence rather than an additional theorem. At the scanned parameters they recover the designed classifier mode counts, locate operational mode-retention boundaries, and show failure through the same two mechanisms indicated by the proof margins: adjacent-clock merger as noise or geometric jitter grows, and suppression of a late mode by depletion as the budget grows. The semi-analytic free clock is consequently a useful preflight diagnostic, but the finite-walker classifier, seed, bandwidth, prominence rule, and time step remain part of the operational definition near a boundary.

The restrictions are substantive. First, the deterministic relative trajectory lies strictly inside the contact region and the free contact probability approaches one in the small-noise limit. The modal skeleton is therefore generated primarily by separated longitudinal exposure clocks, not by a nontrivial contact-gating effect. Second, the extension to $d \geq 2$ uses longitudinal slabs and a transverse symmetry quotient; it demonstrates dimension compatibility rather than a new dimension-driven geometry. Third, the normalized slab height scales as $1/\epsilon$, so the fixed- L^1 constraint imposes neither an independent peak-rate cap nor a minimum feature size. Fourth, B_{top} is a topology threshold, B_{cert} is only a fixed-allocation sufficient value, and B_{op} is a classifier-dependent finite-sample crossing; none is identified with another.

Nothing is proved about topology outside I , a parameter choice uniform in d or m , a common positive event-

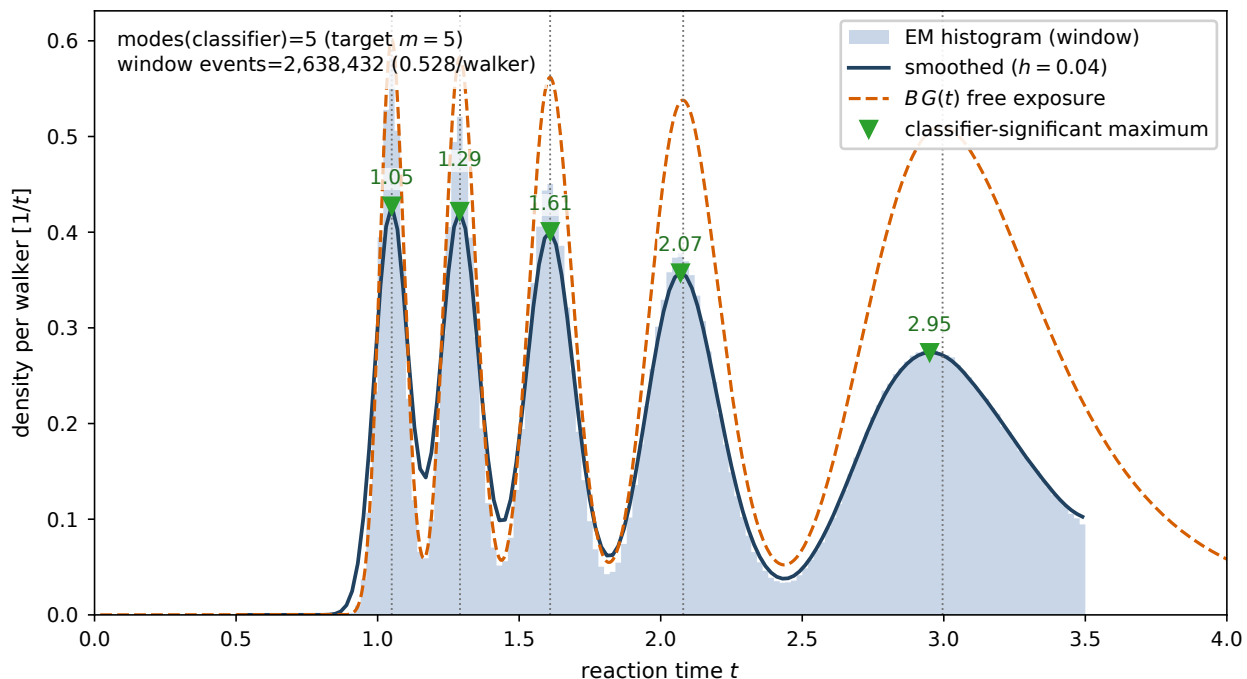


FIG. 4. Five-mode off-lattice realization at $\varepsilon = 0.1$, $B = 1$, uniform weights, 5×10^6 walkers, on a stretched geometry ($z_0 = 8$) with slab centres $\mu_j = (2.8, 2.2, 1.6, 1.0, 0.4)$. The fixed classifier identifies five significant maxima near the design target times (dotted lines), with prominences between 215 and 454 Poisson standard deviations and 0.53 reaction events per walker inside the window. Dashed curve: free-exposure prediction $BG(t)$. (Plotting code revised with Codex assistance; numerical content author-verified.)

mass floor, or mode-count switching on one fixed support family. Natural next questions are whether prescribed modality persists under simultaneous L^1 and L^∞ constraints, under a minimum catalyst width, or when the contact probability remains genuinely nontrivial throughout the design window. Within the stated scope, the result gives a constructive topology theorem for a static resource-constrained Doi field and a finite-parameter test of its mechanism.

ACKNOWLEDGMENTS

The Codex systems and versions identified in the Methods also assisted with literature and cross-document consistency checks and with drafting or revising explanatory

text. Retained citations were checked against the cited records, and the manuscript was rebuilt and inspected after revision. The AI tool is not an author; the sole author takes responsibility for the retained content.

DATA AVAILABILITY

The code, parameter files, machine-readable numerical records, retained histogram counts, classifier diagnostics, figure-generation scripts, and the Lean 4 machine-checked kernel package described in the Supplemental Material are contained in a versioned, checksummed reproduction archive. Its public repository and persistent identifier will be inserted here before portal upload; no experimental data were generated.

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- [26] See Supplemental Material at [URL will be inserted by publisher] for the analytic generator realizations, compact-positive-time mixed-jet bound, complete prescribed-modality proof, fixed-allocation sufficient-budget formula, numerical methods, consistency checks, classifier-sensitivity analysis, configuration tables, and supplementary figures of the off-lattice campaigns.